

Jade Marco S. Morillo

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EDUCATION HISTORY

Far Eastern University – Manila

Nicanor Reyes St., Sampaloc, Manila

Bachelor of Science in Applied Mathematics with Data Science Track

2022-2026

- Entrance scholarship grantee upon admission (2022-2023, 1st Semester)
- Dean's Lister, 1st Honors (2024-2025, 2nd Semester; 2025-2026 1st Semester)
- IAS Parliamentary Examination, Top 5

Manila Tytana Colleges

Diosdado Macapagal Blvd., Pasay

Senior High School Education- Science, Technology, Engineering,
and Mathematics (STEM) strand

2020-2022

- With High Honors (2020-2021; 2021-2022)

St. Dominic College of Asia

Emilio Aguinaldo Hwy., Bacoor,
Cavite

Junior High School Education

2016-2020

- With Honors (2018-2019; 2019-2022)
- Leadership Awardee (2018-2019; 2019-2020)
- With Highest Honors (2017-2018)
- With High Honors (2016-2017)
- Bronze Medalist – Neurolympics (2016-2017)
- Best in Mathematics (2016-2017; 2017-2018)

EXPERIENCES AND ORGANIZATIONS

SMARTech: Social Media Auditing and Reporting Technology – Project Manager & Full-Stack Developer

September 2025 - May 2026

- Architected and deployed an AI-assisted auditing system for the National Youth Commission to automate compliance, branding, and inclusivity verification. The system features a Caption Verifier driven by fine-tuned DeBERTa-v3 and T5 grammar pipelines, alongside a PubMat Checker using a custom YOLOv26 model to automate 9 visual compliance checks.
- Served as the primary liaison for all communications between clients, advisors, and the team to ensure alignment on project goals.
- Led cross-functional team discussions, technical meetings, and milestone presentations.
- Designed and developed the front-end interface and successfully integrated it with back-end APIs and services.
- Contributed to data collection pipelines and assisted with model testing and validation processes.

FEU Central Student Organization Data Analytics Department – Executive Committee

September 2023 - May 2026

- Aggregated, analyzed, and modeled data to generate data-driven reports and proposals that backed university-wide initiatives.
- Engineered and automated Excel-based grading sheets used for the organization's semestral evaluation process, streamlining internal performance tracking.
- Collaborated on large-scale event coordination and provided volunteer leadership for major university events, including One Concierto Piyu, FEU Drag Battle, and Tatak Tamaraw Welcome Fest.

IAS Student Council Marketing and Content Department – Associate Director for Research and Communications

June 2025 - May 2026

- Formulated and executed data-informed social media strategies to maximize audience engagement and increase platform visibility.
- Led the analysis of key performance metrics, translating audience insights into actionable recommendations for content optimization.
- Collaborated on content planning, copywriting, and narrative creation for high-impact online publications.

- Directed cross-departmental communication and alignment among the Office of the Press Relations Offices and other IAS Student Council departments to ensure unified campaign execution.
- Spearheaded departmental operations and logistics for major student council events, including FEU Send-Off 2025 and IAS x IE Sportsfest 2026.

FEU Manila College Y Club Membership Department – Associate Director

June 2025 - May 2026

- Cultivated consistent member engagement by streamlining internal communication channels, broadcasting critical updates, and managing group announcements.
- Maintained membership database integrity, ensuring real-time accuracy, data privacy, and up-to-date tracking of organizational records.
- Served as an official organizational ambassador at prominent internal and external seminars, including TABACOFF Youth Hour 2026 and Addressing the GAP: Gender Affirming Policy Writeshop.

FEU Mathematics Society – Vice President

May 2025 - May 2026

- Directs organizational governance by managing the creation, review, and enforcement of official memorandums, partnership agreements, and policy documents.
- Oversees department operations for both the Legal & Legislative and Student Welfare & Grievances branches, ensuring compliance and student support.
- Conceptualized and launched "Solving Y", a bi-semesteral focus group initiative designed to capture, analyze, and address student grievances regarding academic and university experiences.
- Leads parliamentary participation and internal discussion groups within the IAS Student Council Parliament to advocate for department interests.
- Spearheaded student initiatives and major events, including the Tatak Tamaraw 2025 Program Orientation and the TAMStats 2026 polling event.

A Simulation and Calibration of the Solow-Swan Neo-Classical Growth Model: Analyzing Parameter Sensitivity on the Philippine Economic Trajectory – Project Manager and Co-Author

November 2025 - December 2025

- Co-authored a study analyzing the Philippine Economic Trajectory by implementing Ordinary Least Squares baseline estimations and steady-state calibrations of an augmented Solow-Swan growth model; the framework incorporated fine-tuned tertiary enrollment proxies, institutional quality indicators, and youth dependency structures.
- Formulated data preprocessing and interpolation strategies within Python, engineering the data pipeline to handle missing proxy records, normalize complex institutional datasets, and construct lagged dependent variables for simulation stability.
- Spearheaded the comprehensive data collection pipeline, systematically sourcing and aggregating 40 years of macroeconomic time-series indicators—including GDP per worker, gross investment ratios, and World Governance Indicators—from global and national databases.

Machine Learning Models for Synthesizing and Forecasting the Electricity Outage Risk Index using Time-Series Data – Project Manager and Co-Author

November 2025 - December 2025

- Co-developed a predictive machine learning study forecasting power system vulnerabilities using a hybrid experimental research design; the framework engineered a two-stage cascading architecture utilizing Least Angle Regression (LARS) and LASSO regression to model regional electricity grid risks from multi-domain climate time-series data.
- Conceived and architected the central project methodology by pioneering the concept of a synthesized Electricity Outage Risk Index (EORI) ranging from 1 to 100; this proxy-based framework bypassed severe proprietary data deficiencies and national data infancy in public utility reporting.
- Spearheaded the systematic data collection pipeline, employing Python APIs and web scraping tools to extract, clean, and impute 60 historical data points spanning 15 climate hazard features—including typhoons, sea surface temperature anomalies, and the oceanic Niño index.

GrowTogether: A Collaborative Productivity Application for Gamified Focus and Shared Accountability – Front-End Developer

November 2025 - December 2025

- Co-developed a collaborative, serverless mobile productivity application utilizing Flutter and AWS to promote shared user accountability and gamified focus habits.
- Architected the core front-end interface and mobile UI/UX layouts, engineered interactive component layers, dynamic state-driven navigation grids, and custom data visualizations using Fl Chart.
- Conceived and designed the visual milestone reward architecture, creating a botany-themed awards matrix and multi-category achievement tiers to gamify long-term habit retention and maximize daily streak commitment.

IAS Student Council Documentation Department – Executive Committee

September 2024 - June 2025

- Documented student council events through photography and videography, systematically capturing key program milestones and sponsor activations.
- Produced and optimized multimedia assets for official social media channels, ensuring high-quality visuals across event highlights and promotional campaigns.
- Spearheaded departmental operations and media coverage for FEU SAYAWIT 2025, directing the team to execute real-time content delivery during the university-wide competition.

FEU Manila College Y Club Finance and Logistics Department – Executive Committee

September 2024 - June 2025

- Formulated and reviewed comprehensive budget proposals to optimize resource allocation across all organizational initiatives.
- Streamlined procurement and end-to-end supply chain logistics, managing purchasing pipelines and material safekeeping for major institutional activities.
- Served as a representative, representing both the organization and the university in competitive external events, including the YMCA Olympics and YMCA Sportsfest.

FEU Mathematics Society – Press Relations Officer

May 2024 - May 2025

- Directed multi-channel information dissemination across digital and physical platforms to enhance organizational visibility and audience reach.
- Supervised the organization's creative outputs, establishing oversight on asset development to ensure branding compliance, unified identity, and optimal visual storytelling across all promotional media.
- Oversees the Publicity and Creatives Departments, enforcing brand consistency across the production of all official publications and promotional assets.
- Spearheaded the design, curation, and logistics for the organization's TamHunt promotional booth.
- Led the conceptualization and strategic relaunch of the official FEU Mathematics Society Merchandise Collection, overseeing the project from creative design to campaign release.

IAS Paragon News and Editorial Department – Writer

September 2023 - May 2025

- Authored comprehensive news features and editorial articles covering campus milestones and contemporary socio-political issues for official publication.
- Collaborated closely with other writers to verify fact accuracy, maximize narrative clarity, and maintain strict adherence to institutional journalism guidelines.
- Demonstrated cross-functional agility by actively supporting the Photojournalism Department, capturing supporting visual media to complement written publications.

Business Intelligence Project for Spotify – Data Analyst

September 2024 - December 2024

- Engineered an end-to-end business intelligence solution utilizing Spotify Charts data to track, model, and analyze artist performance and streaming trends.
- Designed and developed interactive Tableau dashboards to visualize complex audience behaviors and key streaming performance metrics.
- Transformed raw streaming datasets into actionable insights, enabling data-driven overviews of seasonal music trends and listener engagement.

SignSense: A Filipino Sign Language Detector – Data Validation Lead and Front-End Developer

September 2024 - December 2024

- Co-developed a real-time computer vision prototype using Python, TensorFlow, Keras, and OpenCV to accurately detect and classify the Filipino Sign Language (FSL) alphabet.
- Designed and engineered the user interface, prioritizing accessibility, intuitive navigation, and clear user guidance for seamless real-time interaction.
- Spearheaded the data validation process, ensuring the integrity, proper labeling, and quality of the image datasets used to train and test the deep learning models.

Kristine's Beauty Salon Website – Front-End Developer

February 2024 - May 2024

- Programmed a dynamic salon web application featuring a live, automated email notification pipeline and an online service booking system implemented through PHP backend scripting.
- Architected the user interface (UI/UX) to prioritize intuitive user navigation, seamlessly reinforcing the salon's corporate branding and visual identity.
- Designed and structured a comprehensive digital catalog to beautifully showcase the salon's full suite of commercial services across fully responsive device layouts.

Wander Pets Website – Assistant Project Manager and Front-End Developer

February 2024 - May 2024

- Co-developed an e-commerce pet retailing platform, engineering critical user-facing features including dynamic state-driven shopping carts and interactive UI components.
- Architected the front-end interface layout, implementing responsive design patterns and intuitive navigation structures to maximize user accessibility.
- Collaborated on product scope management, analyzing essential e-commerce business requirements to ensure comprehensive feature coverage for the digital storefront.
- Assisted in managing the project lifecycle, coordinating development timelines, and aligning design specifications with final application deployment.

FEU Mathematics Society Creatives and Publicity Department – Director

May 2023 - May 2024

- Executed cohesive visual branding guidelines for the FEU Mathematics Society to standardize the organization's digital identity across all public-facing platforms.
- Designed and produced high-quality publicity materials—including promotional posters, digital profile frames, and official certification assets—for university-wide organizational events.
- Collaborated closely with the executive board to translate strategic marketing campaigns into engaging visual assets while maintaining strict aesthetic and production standards.

HASHBlog – Project Manager and Back-End Developer

September 2023 - December 2023

- Co-developed a responsive blogging platform centered on hashtag-based content linkage, engineering a robust backend architecture to support dynamic data relationships.
- Architected and managed an SQL-backed data warehouse, designing relational schemas, writing optimized queries, and ensuring data integrity across user and post records.
- Spearheaded product ideation and feature brainstorming, mapping out core software functionalities based on user-centric requirements.

FEU Mathematics Society – Executive Committee Member

August 2022 – May 2023

- Designed and executed data acquisition strategies by conducting large-scale student surveys during the FEU university elections to gather public opinion, student feedback, and actionable insights.
- Contributed to event planning and survey strategies, ensuring data integrity and representative sampling across various university departments.

Talaria Publication Feature Department – Writer

September 2020 – June 2022

- Authored narrative-driven feature profiles and news articles covering campus milestones, pop culture developments, and contemporary social issues.
- Conducted contextual research and interviews to produce engaging human-interest stories tailored for a diverse student and faculty readership.

SDCA Junior High School Supreme Student Council – Treasurer

March 2019 – March 2020

- Directed the council's administrative operations, engineering strategic budget allocations, and long-term fund reporting frameworks.
- Prepared comprehensive financial statements, systematically auditing revenues, operational costs, and capital expenditures for school-wide events and projects.
- Served as Project Chair for major institutional events, overseeing the end-to-end planning, logistics, and execution of the Valentine's Concert and Booth Day.
- Orchestrated communication with class treasurers, streamlining fund collection pipelines and balancing resource distribution for departmental activities.

SDCA Junior High School Supreme Student Council – Business Manager

March 2018 – March 2019

- Spearheaded fundraising initiatives and commercial operations to generate sustainable revenue for the council's student-led programs and initiatives.
- Enforced organizational policies and financial procedures, auditing class business managers to guarantee strict compliance during major student campaigns.
- Served as Project Co-Chair for high-profile school events, orchestrating end-to-end logistics, resource management, and execution for Mr. & Ms. SDCA and the Christmas Donation Drive.

SDCA Junior High Sipnayan Club – Auditor

September 2017 – March 2018

- Enforced financial transparency and internal controls by systematically monitoring, auditing, and verifying the organization's monetary inflows and outflows.
- Assisted in the end-to-end planning and execution of the club's highly successful Booth Day activity, managing operational logistics, resource deployment, and on-site project workflows.

SKILLS & INTERESTS

- Experienced in end-to-end data science workflows, including data collection, preprocessing, feature engineering, data mining, predictive analytics, statistical modeling, and machine learning
- Skilled in artificial intelligence and machine learning development using Python, TensorFlow, OpenCV, YOLOv8, DeBERTa-v3, T5, and regression-based modeling techniques
- Experienced in time-series forecasting, econometric modeling, simulation, calibration, sensitivity analysis, multivariate analysis, and quantitative research methodologies
- Strong foundation in calculus, linear algebra, differential equations, numerical analysis, probability, linear models, mathematical modeling, and quantitative reasoning
- Knowledgeable in management science, operations research, fundamentals of accounting, and theory of interest
- Skilled in data mining, web scraping, API utilization, data cleaning, imputation, and large-scale data pipeline engineering
- Strong background in business intelligence, dashboard development, statistical reporting, Excel automation, and data visualization using Tableau and interactive reporting tools
- Knowledgeable in database management, relational schema design, SQL query optimization, data warehousing, and cloud computing concepts
- Proficient in Python, SQL, R, C++, Git, Tableau, LaTeX, Visual Studio Code, Microsoft Office 365, Google Workspace, and other productivity and collaboration tools
- Experienced in full-stack and front-end development using Flutter, PHP, SQL, AWS, API integration, and responsive UI/UX design principles
- Passionate about artificial intelligence, automation systems, predictive analytics, economic modeling, climate risk analytics, and socially impactful technology
- Skilled in technical writing, research documentation, academic writing, and data storytelling for data-driven communication and decision-making
- Experienced in social media analytics, audience engagement analysis, content strategy, and performance-driven digital communication
- Skilled in copywriting, editorial writing, branding, multimedia production, photography, videography, graphic design, and visual storytelling
- Strong analytical thinking, critical reasoning, and problem-solving abilities developed through technical research, systems development, and organizational leadership
- Demonstrated leadership in organizational governance, policy drafting, event operations, logistics management, and volunteer coordination
- Effective in client relations, professional correspondence, public presentations, teamwork, and collaborative decision-making
- Adaptable, detail-oriented, and highly organized with the ability to manage multiple projects and responsibilities efficiently in fast-paced environments
- Demonstrates initiative, accountability, professionalism, and a strong commitment to continuous learning and innovation

PARTICIPATIONS AND SEMINARS

Proponentship

- FEU Mathematics Society TAMStats 2026: Connecting the Pieces through Unbounded Polling, Proponent and Officer-in-Charge
- PROJECT BOOST 2025: Quantitative Data Analysis Workshop Seminar, Proponent
- IAS Student Council IAS-On-The-Scene, Co-Proponent and Scriptwriter
- FEU Mathematics Society Merchandise Collection, Proponent and Officer-in-Charge

- SDCA Mr. and Ms. Teen SDCA, Co-Proponent
- SDCA Supreme Student Council-JHS Christmas Donation Drive, Co-Proponent
- SDCA JHS PebRock 2020, Proponent
- SDCA JHS Booth Day, Proponent

Officer-in-Charge, Head, and/or Working Committee

- 9th IAS Parliament, Academic Committee Member
- IAS x IE Sportsfest 2026, Marketing and Content Head
- FEU Drag Battle 2026, Runners Working Committee
- Liberal Youth Political Forum on Liberalism and Governance, Marketing and Content Working Committee
- FEU Mathematics Society September Discussion Group, Officer-in-Charge
- FEU Mathematics Society TamHunt2025: Weave Got the X, Booth Design and Operations Working Committee, and Social Media PR head
- FEU College Y Club TamHunt 2025: CYCibol sa Himpampawid, Marketing and Content Working Committee
- IAS Student Council TamHunt 2025: Cubao IASpo, Marketing and Content Working Committee
- IAS Student Council Cubao IASpo: Short Film Series, Creative Director
- AI and the Future of Work: 2025-2035, Working Committee
- FEU Mathematics Department General Orientation Day, Technical Head
- IAS Student Council IAS Studios, Creative Director
- FEU Mathematics Society Tatak Tamaraw 2025 Program Orientation, Officer-in-Charge
- IAS Student Council Tatak Tamaraw 2025 Plenary Session, Marketing and Content Head
- IAS Student Council Send-Off 2025, Marketing and Content Co-Head
- FEU Mathematics Society Send-Off 2025, Officer-in-Charge
- 2025 Ready, Set, Work: A Pre-employment Initiative of the IAS Student Council, Marketing and Content Co-Head
- FEU College Y Club Send-Off 2025, Technical Head
- 2025 FEU Townhall Meeting on Blended Modality, Production and Technicals Working Committee
- FEU Mathematics Society Midyear Semester Discussion Group, Officer-in-Charge
- FEU Mathematics Society TAMStats 2025: Projecting Victory through Data and Discourse, Creatives and Publicity Head, and Data Analytics Working Committee
- IAS Paragon TAMStats 2025 Launching Event, Photojournal Working Committee
- FEU Mathematics Society TAMStats 2025 Launching Event, Events, Technicals, and Documentation Working Committee
- IAS Student Council The Living Heritage Student Congress 2025, Production and Technicals Working Committee
- FEU Athletics League 2025 IAS Basketball Team, Documentation Working Committee
- FEU SayAwit 2025 IAS Sessionistas and Dance Company, Documentation Co-Head
- IAS Sportsfest 2025: The Rise of Districts, Documentation Working Committee
- IAS Sportsfest 2025 FEU Mathematics Society, Officer-in-Charge
- One Concierto Piyu 2025, Refreshments Working Committee
- FEU Drag Battle 2025, Registration Working Committee
- IAS Paragon Valentine Special, Photojournal Working Committee
- Quezon City Science High School Work Immersion in FEU, Officer-in-Charge
- IAS Student Council BOOST Your Excel Skills, Documentation Working Committee
- FEU Mathematics Society Skill Up with LaTeX: A Workshop Seminar, Officer-in-Charge
- FEU Mathematics Society TamFeels, Officer-in-Charge
- FEU Mathematics Society TamHunt 2024: My X and Ys Booth, Officer-in-Charge and Social Media PR and Booth Design Head
- FEU Central Student Organization TamHunt 2024: FEUCSO TV, Volunteer
- FEU Mathematics Society Tatak Tamaraw 2024 Program Orientation, Usher Head and Officer-in-Charge
- IAS Student Council Tatak Tamaraw 2024 Plenary Session, Logistics Working Committee
- FEU Central Student Organization Tatak Tamaraw 2024 Tams Challenge, Runner
- FEU Central Student Organization Tatak Tamaraw 2024 Tatakan Rights, Marshal
- FEU Central Student Organization Tatak Tamaraw 2024 Welcome Fest, Marshal
- FEU Mathematics Society Send-Off 2024, Officer-in-Charge
- FEU Mathematics Society Preview 2024, Officer-in-Charge
- FEU Mathematics Society TamHunt 2023 M's Pisonet, Creatives and Publicity Co-Head
- FEU Mathematics Society TAMStats 2023, Working Committee

Writing

- IAS x IE Sportsfest 2026, Scriptwriter

- 9th IAS Parliament Parliament Patrol, Scriptwriter
- IAS Student Council IAS on the Scene, Scriptwriter
- IAS Student Council IAS Studios, Scriptwriter
- IAS Paragon Newsletter 2025, Writer
- IAS Paragon Reconnecting through Play, Writer
- IAS Paragon Glowing Green and Gold: Pasko sa PIYU 2024, Writer
- IAS Paragon Christmas Season: The Filipino Time Exception, Writer
- IAS Paragon Sakripisyo'y Suklian, Atletang Ating Ipaglaban, Writer
- IAS Paragon EDSA People Power Erasure: An Attempt to Revise History, Writer
- IAS Paragon No More Treats?: The Horror of Inflation in the Philippines, Writer
- Talaria Publications Elections 2022, Writer
- Talaria Publications Women's Month 2022, Writer
- Talaria Publications National Reading Month 2021, Writer
- Talaria Publication OPM Feature, Writer

Seminars and Other Participation

- TechEverywhere 2025: ReAIImagined, Participant
- Radical Compassion: Restoring Dignity Through Second Chances, Attendee
- Addressing the GAP: Gender Affirming Policy Writeshop, Participant
- DOLE Labor Education for Graduating Students, Participant
- How to Stand Out as a Graduate: Your guide to landing the job that's right for you with Prosple, Participant
- IAS Student Council 2025 First Ka! The Rise of Student-First Leadership Summit, Participant
- FEU Academy x NRMF Training Series session on Digital Marketing and Fundraising for Impact, Participant
- TobacOFF Youthour 2025: Take Back Our Future, Participant
- TobacOFF TOURopa Caravan FEU, Participant
- FEU Center for Learning Enrichment and Research for Students Critical Thinking Workshop, Participant
- FEU Student Development SOAR Discussion, Participant
- FEU Center for Learning Enrichment and Research for Students Productivity Workshop, Participant